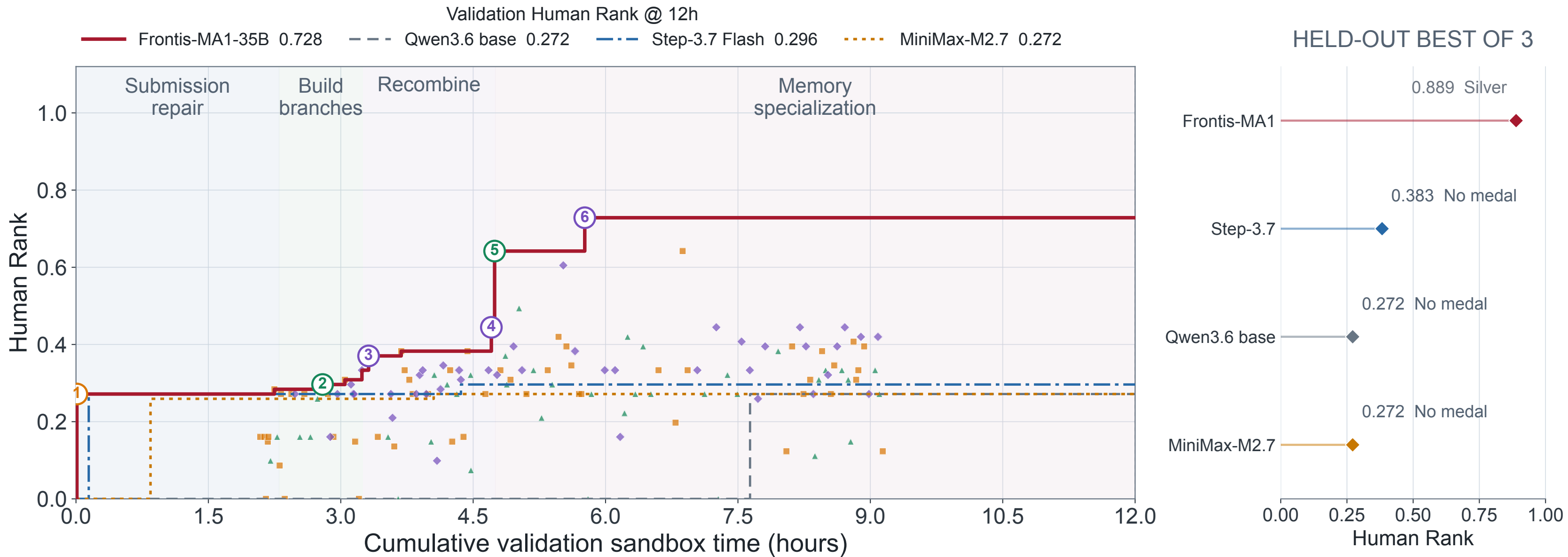


Repair and Recombination Produce a Silver Bird Detector

mlsp-2013-birds • epoch 0 • OpenMLE-Evo • Frontis-MA1 nodes with matched baselines



Frontis-MA1 Milestones

Key program changes at the numbered trajectory nodes.

● Draft ■ Debug ▲ Improve ◆ Crossover

1 DEBUG: VALID SUBMISSION

Fixes the sample-submission merge and stabilizes LightGBM under severe class imbalance.

step 5 AUC 0.74786 Human Rank 0.2716

2 IMPROVE: AUDIO FUSION

Fuses filtered spectrograms and segment histograms with EfficientNet-B0 and an MLP.

step 48 AUC 0.79390 Human Rank 0.2963

3 CROSSOVER: ROBUST AUDIO

Combines safe parsing, SpecAugment, stratified CV, and EfficientNet-B2 with TTA.

steps 71-72 AUC 0.82774 Human Rank 0.3704

4 CROSSOVER: CLASS BALANCE

Fuses aligned multimodal features with safe per-class AUC and class weighting.

step 118 AUC 0.85744 Human Rank 0.4444

5 IMPROVE: FOCAL + TTA

Uses focal loss and TTA to improve rare-species predictions.

step 119 AUC 0.87737 Human Rank 0.6420

6 CROSSOVER: MEMORY WINNER

Selects the robust EfficientNet-B2 branch with SpecAugment, OneCycleLR, and safe K-fold handling.

step 150 AUC 0.88576 Human Rank 0.7284